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Application of Power Electronics for Renewable Generation - Assignment 2
Report

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1 Task 2.1: Implementation of the Photovoltaic (PV) Array Model

1.1 Introduction

Photovoltaic (PV) arrays consist of many interconnected solar cells or modules that convert sunlight directly into electrical energy via the photovoltaic effect [1]. In practice, individual PV cells (typically made of crystalline silicon) are assembled into panels, and panels are connected in series and parallel to form larger arrays [1]. The electrical behavior of a PV array is governed by several key parameters. Solar irradiance (the incident sunlight power per unit area) directly drives the generated current: higher irradiance leads to proportionally higher output current and power [2]. In contrast, elevated cell temperature (due to ambient heat or self-heating under sun) causes the open-circuit voltage of the modules to drop, reducing overall efficiency [3]. Finally, PV arrays exhibit non-linear current–voltage (I–V) curves with a distinct maximum power point that shifts with irradiance and temperature.

1.2 Photovoltaic (PV) Array Model

A detailed model of a photovoltaic (PV) array is outlined below. Initially, based on the PV module data, the diode's leakage current I_0 and the current generated by incident solar radiation I_{pv} can be determined, considering different temperature and irradiance values.

$$V_t = N_{s,\text{cell}} \cdot \frac{k \cdot T}{q} \quad (1)$$

$$I_0 = \frac{I_{sc,n} + K_I(T - T_n)}{\exp\left(\frac{V_{oc,n} + K_V(T - T_n)}{V_t \cdot a}\right) - 1} \quad (2)$$

$$I_{pv,n} = I_{sc,n} \cdot \frac{R_p + R_s}{R_p} \quad (3)$$

$$I_{pv} = [I_{pv,n} + K_I(T - T_n)] \cdot \frac{G}{G_n} \quad (4)$$

If a resistive load is connected, the relationship between the PV array voltage V_a and current I_a is given by:

$$V_a = R \cdot I_a \quad (5)$$

The operating point of the PV array is determined using the single PV module model. The current of an individual module can be expressed as:

$$I_m = I_{pv} - I_0 \left(\exp\left(\frac{V_m + R_s \cdot I_m}{V_t \cdot a}\right) - 1 \right) - \frac{V_m + R_s \cdot I_m}{R_p} \quad (6)$$

where V_m and I_m represent the voltage and current of a PV module, respectively.

For a complete PV array, the total array voltage and current relate to the module-level values as:

$$V_p = V_a = V_m \cdot N_s; \quad I_p = I_a = I_m \cdot N_p \quad (7)$$

Since V_a and I_a are both unknowns, one must be substituted using the relationship given in (5). This simplifies equation (8) to involve only a single unknown. If $V_a = I_a \cdot R$, then equation (8) becomes:

$$I_a = N_p \cdot I_{pv} - N_p \cdot I_0 \left(\exp \left(\frac{V_a/N_s + R_s \cdot I_a/N_p}{V_t \cdot a} \right) - 1 \right) - N_p \left(\frac{V_a/N_s + R_s \cdot I_a/N_p}{R_p} \right) \quad (8)$$

This nonlinear equation is solved using an iterative approach. In MATLAB, the `fsolve` function can be employed to find the solution. An initial guess for the current, $I_{m,0}$, is necessary to start the iteration process; for example, $I_{m,0} = I_{sc,n}(T - T_n)$. Alternatively, equation (8) for the PV array can be written as:

$$I_a = N_p \cdot I_{pv} - N_p \cdot I_0 \left(\exp \left(\frac{I_a (R/N_s + R_s/N_p)}{V_t \cdot a} \right) - 1 \right) - N_p \left(\frac{I_a (R/N_s + R_s/N_p)}{R_p} \right) \quad (9)$$

The PV array model is shown in Fig. 1

Photovoltaic generation - Modeling

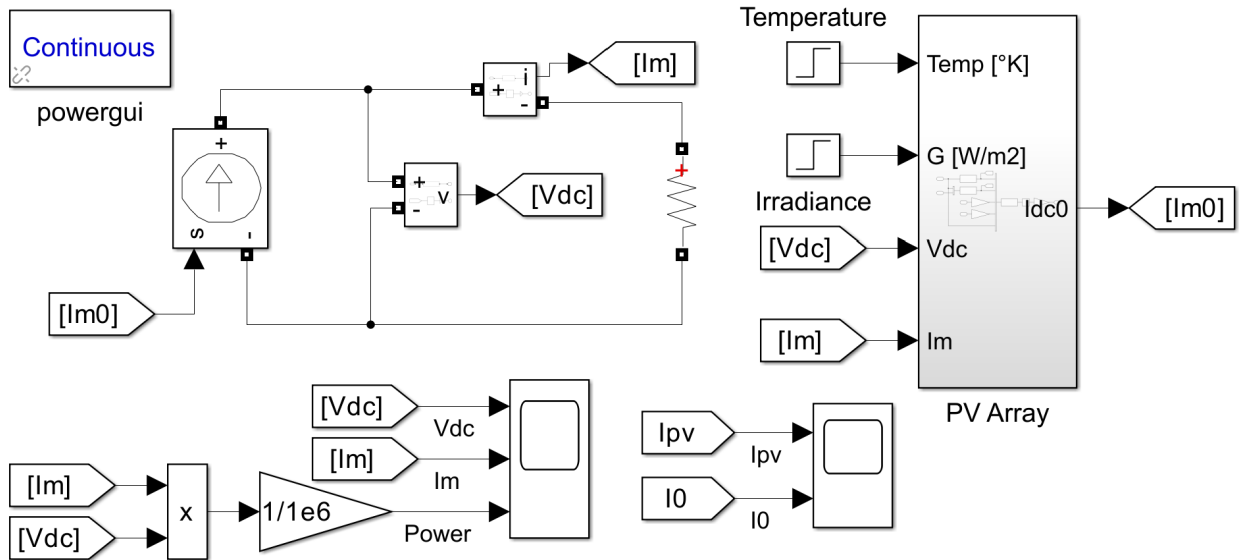


Figure 1: PV array model

1.3 Impact of Temperature (T) and Irradiance (G) Variations on PV Current and Voltage

Photovoltaic modules are highly sensitive to changes in environmental conditions, particularly temperature (T) and irradiance (G), which significantly influence their electrical output characteristics.

As shown in (6), both irradiance and temperature directly affect the current generated by the PV cells. An increase in irradiance leads to a higher photocurrent I_{pv} . In contrast, lower temperatures improve voltage levels but slightly reduce current. These dependencies underline the importance of accurately modeling both temperature and irradiance in dynamic PV system simulations.

The dynamic behavior of the PV current and voltage under varying temperature and irradiance conditions is illustrated in Fig. 2. Three types of scenarios were tested to analyze the response of key parameters including V_{dc} , I_m , Power, I_{pv} , and I_0 , as described below:

- **Constant Irradiance, Decreasing Temperature:**

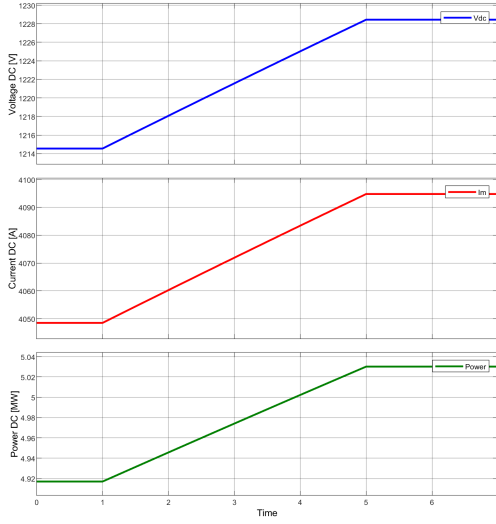
Fig. 2a and Fig. 2b present the results when the irradiance G is held constant at 800 W/m^2 , while the temperature decreases progressively by 1°C every second from second 1 to second 6, ranging from 25°C down to 21°C . Under these conditions, the voltage V_p , current I_p , and power increase as the temperature decreases. Meanwhile, both I_{pv} and I_0 exhibit a decreasing trend.

- **Constant Temperature, Decreasing Irradiance:**

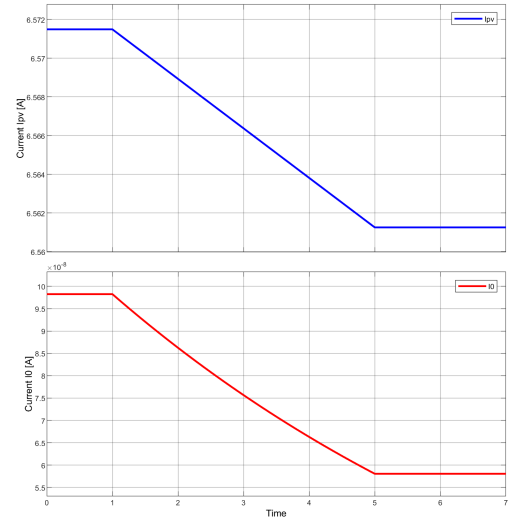
Fig. 2c and Fig. 2d show the results when the temperature is fixed at 25°C and the irradiance decreases progressively every second from second 1 to second 6, ranging from 800 to 600 W/m^2 in steps of 50 W/m^2 . In this scenario, all parameters V_p , I_p , power, and I_{pv} decrease with irradiance. However, I_0 remains constant, as it depends solely on temperature, in accordance with (2).

- **Decreasing Both Irradiance and Temperature:**

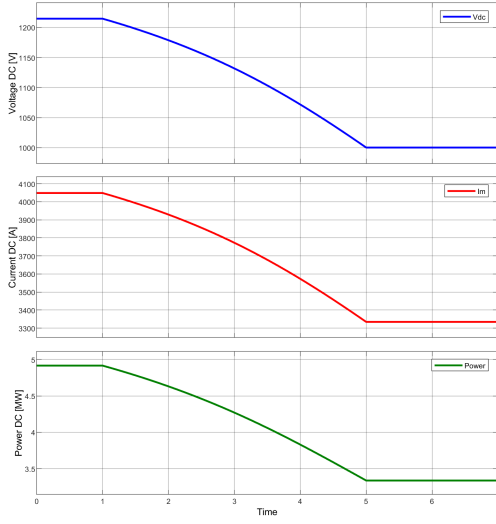
Fig. 2e and Fig. 2f depict the case where both irradiance and temperature are decreasing simultaneously using the same values as above. In this combined scenario, all parameters including I_0 decrease over time, reflecting the compounded effect of lower irradiance and temperature.



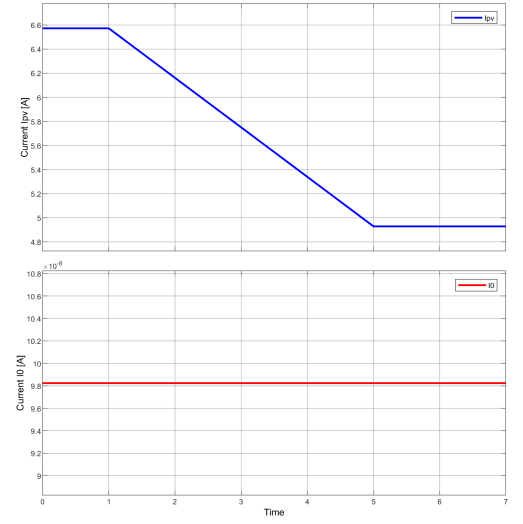
(a) V_{dc} , I_m , and Power at $G = 800 \text{ W/m}^2$ and different T



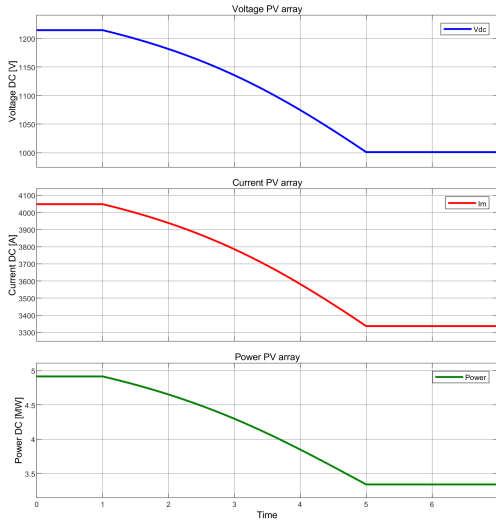
(b) I_{pv} and I_0 at $G = 800 \text{ W/m}^2$ and different T



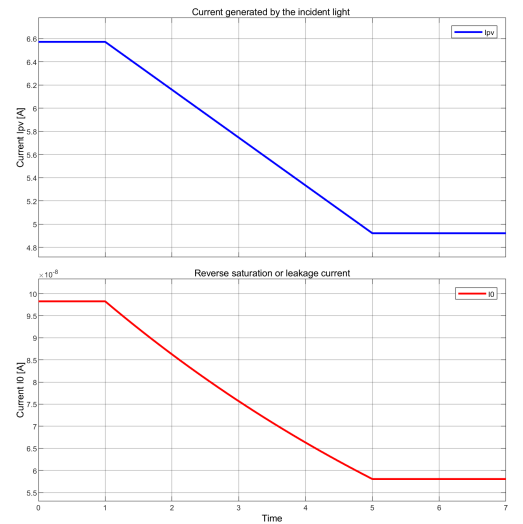
(c) V_{dc} , I_m , and Power at different G and $T = 25^\circ\text{C}$



(d) I_{pv} and I_0 at different G and $T = 25^\circ\text{C}$



(e) V_{dc} , I_m , and Power at different G and T



(f) I_{pv} and I_0 at different G and T

Figure 2: Dynamic behavior of the PV array under different irradiance (G) and temperature (T) conditions.

1.4 Validation of Steady-State Simulation Results

To ensure the accuracy of the developed PV array model, a validation process was conducted by comparing the dynamic simulation results obtained from Simulink with the steady-state calculations performed in MATLAB.

The validation covered three distinct operating scenarios: (1) constant irradiance with decreasing temperature, (2) constant temperature with decreasing irradiance, and (3) simultaneous variation of both temperature and irradiance. In each case, key electrical parameters demonstrated strong agreement between the dynamic simulation (Fig. 2) and the steady-state results (Fig. 3). This consistency confirms the correctness of the model and supports its use for further analysis, including control design, performance evaluation, and integration studies within more complex power systems.

	I _{pv} [A]	I ₀ [A]	Voltage V _p [V]	Current I _p [kA]	Power [MW]
G = 800 W/m², T = 25 °C	6.5715	9.8252e-08	1214.5	4.0485	4.9171
(a) Irradiance $G = 800 \text{ W/m}^2$ and Temperature $T = 25^\circ\text{C}$					
	I _{pv} [A]	I ₀ [A]	Voltage V _p [V]	Current I _p [kA]	Power [MW]
G = 800 W/m², T = 24 °C	6.5689	8.6257e-08	1218.1	4.0603	4.9457
G = 800 W/m², T = 23 °C	6.5664	7.566e-08	1221.6	4.0719	4.9741
G = 800 W/m², T = 22 °C	6.5638	6.6307e-08	1225	4.0834	5.0023
G = 800 W/m², T = 21 °C	6.5613	5.8057e-08	1228.4	4.0948	5.0302
(b) Irradiance $G = 800 \text{ W/m}^2$ at Different Temperatures T					
	I _{pv} [A]	I ₀ [A]	Voltage V _p [V]	Current I _p [kA]	Power [MW]
G = 750 W/m², T = 25 °C	6.1608	9.8252e-08	1178.7	3.929	4.6312
G = 700 W/m², T = 25 °C	5.7501	9.8252e-08	1131.9	3.7729	4.2704
G = 650 W/m², T = 25 °C	5.3393	9.8252e-08	1071.9	3.5731	3.8301
G = 600 W/m², T = 25 °C	4.9286	9.8252e-08	1000.3	3.3344	3.3356
(c) Different Irradiance G and Temperature $T = 25^\circ\text{C}$					
	I _{pv} [A]	I ₀ [A]	Voltage V _p [V]	Current I _p [kA]	Power [MW]
G = 750 W/m², T = 24 °C	6.1584	8.6257e-08	1181.6	3.9386	4.6538
G = 700 W/m², T = 23 °C	5.7456	7.566e-08	1135.7	3.7857	4.2994
G = 650 W/m², T = 22 °C	5.3331	6.6307e-08	1074.7	3.5822	3.8497
G = 600 W/m², T = 21 °C	4.9209	5.8057e-08	1001.1	3.3371	3.3408
(d) Different Irradiance G and Temperature T					

Figure 3: Steady-state PV array results under different irradiance (G) and temperature (T) conditions

2 Task 2.2: Implement DC Voltage Control

2.1 DC voltage control

The DC voltage control is implemented to regulate the active power in order to maintain the DC bus voltage at its desired value. This controller generates the reference signal i_q^* for the inner current control loop. The DC voltage loop must be slower than the inner current control loop otherwise it will cause instability problem in cascaded control structure. [4]

The DC voltage controller is designed based on the following transfer function 10.

$$V_{dc}(s) = \frac{G_v(s)}{C_{dc}s + G_v(s)} V_{dc}^*(s) + \frac{1}{C_{dc}s + G_v(s)} (I_{dc-m}(s)) \quad (10)$$

The closed-loop system is designed to behave like a standard second order system in equation 11. It is characterized by the natural frequency ω_n and the damping ratio ξ .

$$\frac{I_{dc-m}}{V_{dc}} = \frac{1}{C_{dc}s + G_v(s)} = \frac{s}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad (11)$$

The DC voltage controller is implemented as a Proportional-Integral (PI) controller to avoid the steady state error that caused by the disturbance:

$$G_v(s) = k_{pv} + \frac{k_{iv}}{s} \quad (12)$$

Thus, the transfer function of the system becomes:

$$\frac{I_{dc-m}}{V_{dc}} = \frac{s}{C_{dc}s^2 + k_{pv}s + k_{iv}} \quad (13)$$

By defining a desired natural frequency ω_n and damping ratio ξ for the closed-loop system, the controller gains k_{pv} and k_{iv} can be determined:

$$k_{pv} = 2\xi\omega_n C_{dc} \quad (14)$$

$$k_{iv} = \omega_n^2 C_{dc} \quad (15)$$

2.2 Tuning of the control

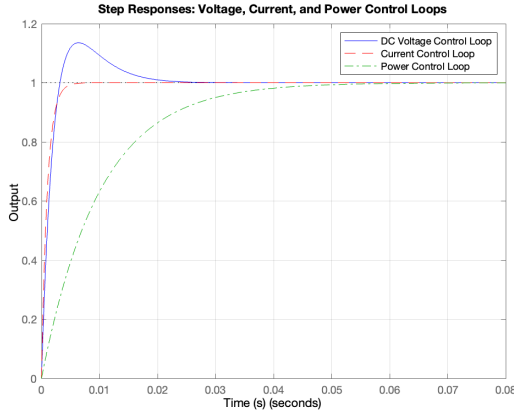
Tuning k_{pv} and k_{iv} through ω_n and ξ directly influences the dynamic behavior of the closed-loop system. The natural frequency ω_n governs the speed of the system response, where a higher ω_n results in a faster transient response and quicker regulation of the DC voltage.

2.2.1 Tuning of ω_n

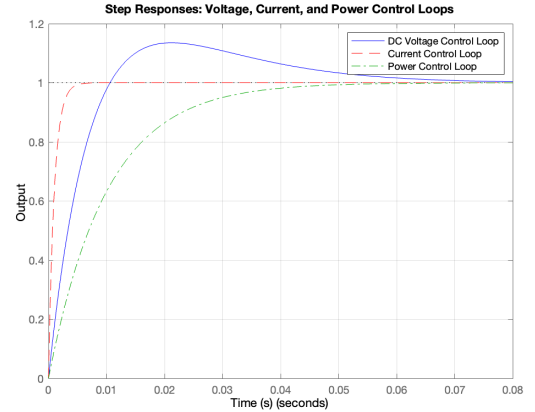
Previously in the VSC controller design, a power control loop was implemented as the outer-most loop, intentionally tuned to be at least one decade slower than the inner current control loop. In this task, the outer loop is replaced by a DC voltage control loop, which now requires careful tuning of its proportional and integral gains (K_p and K_i). To guide the tuning process, the prior power loop response is used as a reference.

To analyze the impact of different tuning strategies, two scenarios are considered based on the natural frequency ω_n and a damping ratio $\zeta = 1$ (critically damped). Specifically, the tested natural frequencies are $\omega_n = 300$ rad/s and $\omega_n = 100$ rad/s. The transfer function plotted for the DC voltage control loop corresponds to equation 16, excluding the $I_{dc-m}(s)$ term, in order to evaluate the response of $V_{dc}(s)$ to the reference input $V_{dc}^*(s)$.

$$\frac{V_{dc}(s)}{V_{dc}^*(s)} = \frac{G_v(s)}{C_{dc}s + G_v(s)} \quad (16)$$



(a) Step response for $\omega_n = 300$



(b) Step response for $\omega_n = 100$

Figure 4: Comparison of step responses for different natural frequencies.

The step response for $\omega_n = 300$, shown in Figure 4a, exhibits significantly faster dynamics than the current control loop in the rising time, indicating that this setting is too aggressive.

In contrast, the response shown in Figure 4b for $\omega_n = 100$ achieves a more desirable trade-off. It settles noticeably faster than the original power loop while remaining sufficiently slower than

the current loop. This places the DC voltage control dynamics within an appropriate intermediate range.

Another approach to justify the selection is by estimating the time constant using $\tau \approx 1/\omega_n$ and comparing it with the time constant of the current control loop (approximately 1 ms):

- For $\omega_n = 300$, the estimated time constant is approximately 3.33 ms, which is 3.3 times slower than the current loop.
- For $\omega_n = 100$, the estimated time constant is around 10 ms, aligning with the previous power loop (a decade slower than the current loop) and maintaining the intended time-scale separation.

This comparative approach aids in selecting value of ω_n for the DC voltage loop.

2.2.2 Tuning of ξ

The step response shown in Fig. 5 corresponds to the transfer function in Equation 13. With a fixed natural frequency ω_n , different values of the damping ratio ζ (0.8, 1, and 2) are plotted to illustrate their effects on system behavior. Specifically, $\zeta = 0.8$ represents an underdamped response, $\zeta = 1$ a critically damped response, and $\zeta = 2$ an overdamped response.

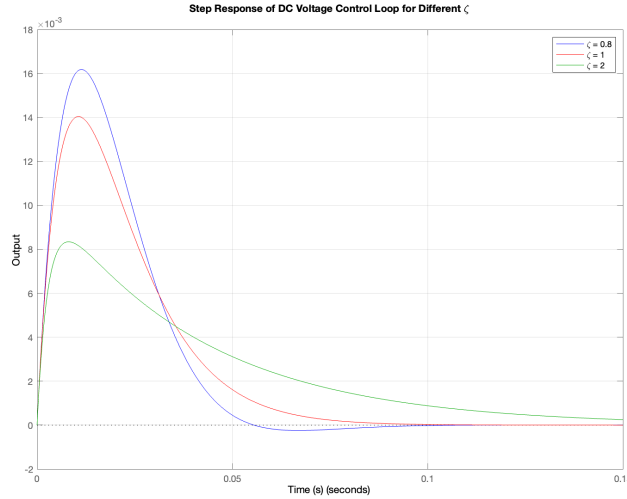


Figure 5: Step response for different damping ratios $\zeta = 0.8, 1$, and 2 .

To approximate a first-order system and avoid overshoot, it is desirable for the response to be sufficiently damped. In an overdamped system ($\zeta > 1$), the two poles are real and distinct, resulting in a response that is the sum of two decaying exponentials. These are characterized by time constants τ_1 and τ_2 , given by equation 17 [5].

$$\tau_1 = \frac{1}{\left(\zeta + \sqrt{\zeta^2 - 1}\right) \omega_n}, \quad \tau_2 = \frac{1}{\left(\zeta - \sqrt{\zeta^2 - 1}\right) \omega_n} \quad (17)$$

When ζ is large, $\tau_1 \gg \tau_2$, meaning the faster exponential decay governed by τ_2 quickly vanishes, and the system response becomes dominated by the slower term τ_1 . This causes the overall system behavior to resemble that of a first-order response.

However, increasing ζ also makes the response slower overall, which may reduce the system's ability to reject disturbances efficiently. Therefore, there is a trade-off between minimizing overshoot and achieving good disturbance rejection.

2.3 The effect of external DC current injection

The changes in the injected DC current I_{dc} can cause fluctuations in the DC bus voltage. To mitigate this, a feedforward control strategy is introduced to improve disturbance rejection. The feedforward term uses the measured disturbance I_{dc} to directly adjust the controller output, thereby counteracting the disturbance before it significantly impacts the DC voltage. This predictive action enhances the system's dynamic performance by reducing the effect of external disturbances in real time. Without the feedforward compensation, the proportional-integral (PI) controller would rely solely on feedback to react after the disturbance has already affected the system, resulting in larger voltage deviations.

2.4 Simulation Results

To test the constructed V_{DC} controller, the converter model is subjected to a sudden increase in the input current. The physical significance of such a disturbance is a change in the PV current that might happen due to changing irradiance or other externalities. The input current is stepped at 0.2 seconds from 0 to 4 kA. Since the current at maximum irradiance of this same array was 4048.1 A from the PV MATLAB exercise, this step size is an estimate the largest that the converter could experience, ostensibly by turning on the converter in the middle of a sunny day. This is an extreme case, since the input current is ultimately determined by temperature and irradiance, and these two quantities change quite slowly in reality. The response of the converter V_{DC} is examined in the next sections with different PI parameters and feed-forward values.

2.4.1 Feed-forward Included

The first example is shown below in Fig. 6. The controller is tuned to be underdamped with a natural frequency of 500 rad/s, and includes the feedforward current after the control output.

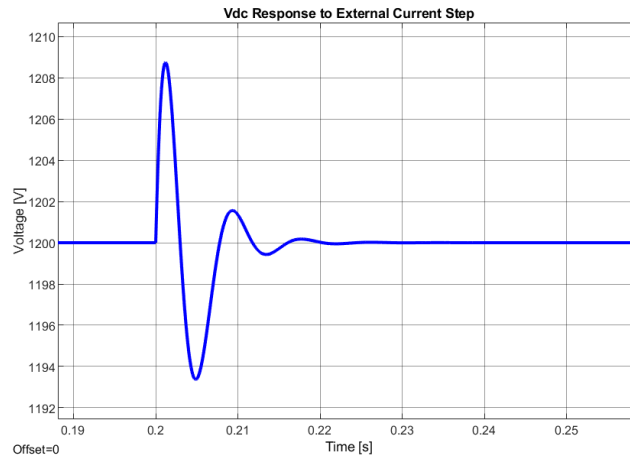


Figure 6: Voltage Response with $\xi = 0.8$ and $\omega = 500$

The response is quite fast and oscillatory, which indicates that the natural frequency is relatively high and that the controller should have more damping. In the next two simulations, the frequency is lowered and both a critically damped and overdamped response are tested.

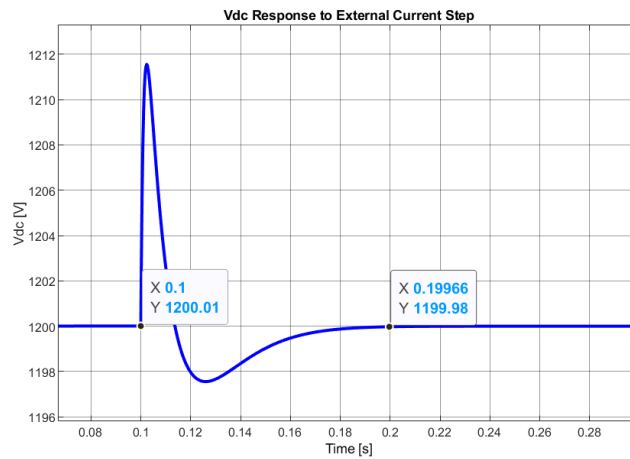


Figure 7: Voltage Response with $\xi = 1$ and $\omega = 75$

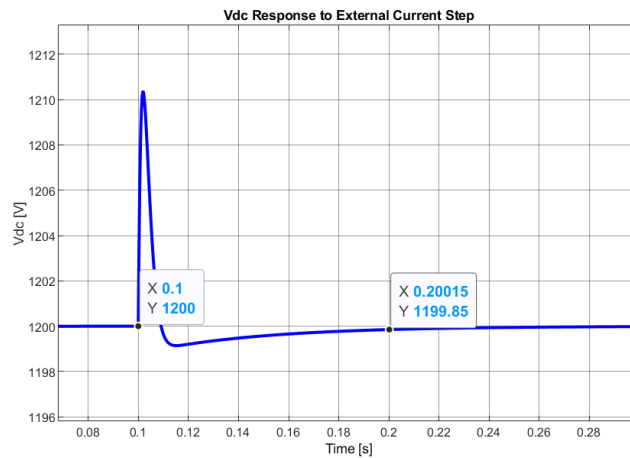


Figure 8: Voltage Response with $\xi = 2$ and $\omega = 75$

The decreased frequency has slowed down the response considerably and eliminated most of

the oscillations, which is expected since the natural frequency is related directly to the proportional to the proportional term and quadratically related to the integral term. Reducing this significantly will reduce both PI terms and therefore the response speed.

The damping constant is related only the proportional constant. Increasing it while keeping the frequency constant decreased the overshoot as visible in Figs. 7 and 8. Therefore, in order to achieve a response that is both fast and approximates a first-order response (that is, without any overshoot), a small frequency and a high damping ratio could be combined, as in Fig. 9.

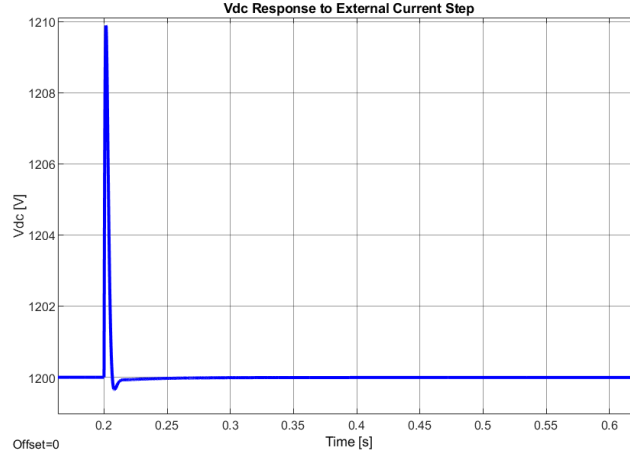


Figure 9: Voltage Response with $\xi = 22$ and $\omega = 10$

This appears to achieve the desired response quite quickly, but this tuning leaves the integral constant too low to remove constant errors, shown in detail in Fig. 10.

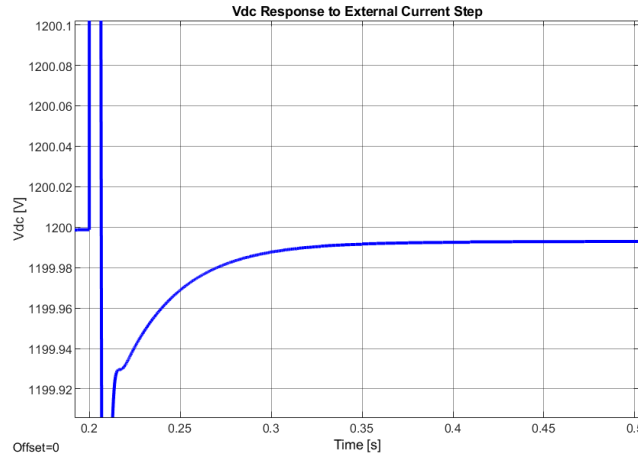
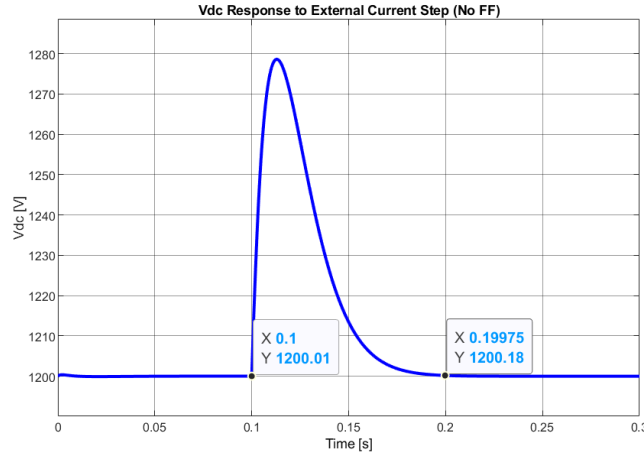


Figure 10: Voltage Response with $\xi = 22$ and $\omega = 10$

Clearly, then, the best response appears to be a compromise between minimizing overshoot and minimizing response time, such as the critically damped response in Fig. 7.

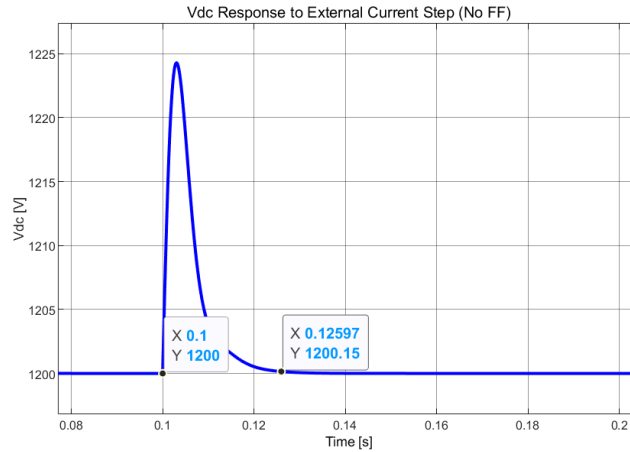
2.4.2 Feed-forward Excluded

Next, the current feed-forward term is removed from the controller and its behavior examined. The best controller from last section Fig. 11 is simulated below.


 Figure 11: Voltage Response with $\xi = 1$ and $\omega = 75$

The same current disturbance clearly produces a much higher voltage increase when the feed-forward term is excluded. It is also the cause of the overshoot in the critically damped response, which normally does not overshoot. The feed-forward term adds to the controller's response regardless of the actual voltage error, which has two effects. It helps the controller respond without allowing the error to rise as much, which results in the voltage spikes from the current disturbance being much lower. The term also contributes to the controller overshooting the reference, since it adds to the control output even after the voltage error is quite small, undermining the proportional controller.

In order to decrease the response time without the feed-forward, the natural frequency is increased significantly. The tuned response is shown below in Fig. 12.


 Figure 12: Voltage Response with $\xi = 1$ and $\omega = 300$

Both models were tuned for the fastest response time while still approximating first-order shape. The results are shown in Figs. 7 and 12, with the simulation excluding the feed-forward term exhibiting a much faster response. When excluding the feed-forward term, the PI constants can be increased further without adding overshoot and oscillations. The trade-off then is a faster response for a higher disturbance peak. The feed-forward term is included in the design because the lower peak is preferable for this application than a fast response. When the DC

controller is too quick, it interferes with the current loop while setting the i_q reference without being sufficiently delayed.

3 Task 2.3: Complete PV Array and VSC Model

Now that the DC voltage controller is tuned, the VSC model can be further completed by combining it with the PV array model from Task 2.1. Instead of a direct current source feeding the DC side of the VSC, the current will be calculated by the PV array according to inputs of irradiance and temperature. The response of the system will therefore be examined under changes in these quantities.

3.1 Irradiance and Temperature Variation

To examine the response of the integrated PV block, the temperature and irradiance are varied in the same way as in Section 1.4 to be able to compare the simulation steady-state with the results calculated in MATLAB. The measurements shown for each graph in the following subsections can be validated with the table in Sections 1.4 or 3.3.

Since the V_{DC} reference is set manually for now, it will be set to the value expected by the calculation. If the current and power also match that calculation, then the simulation is validated.

3.1.1 Base Case

First the base case is simulated, with a nominal temperature of 25°C and irradiance of 800 W/m². The V_{DC} reference is set to 1214.5 V in accordance with Fig. 3a.

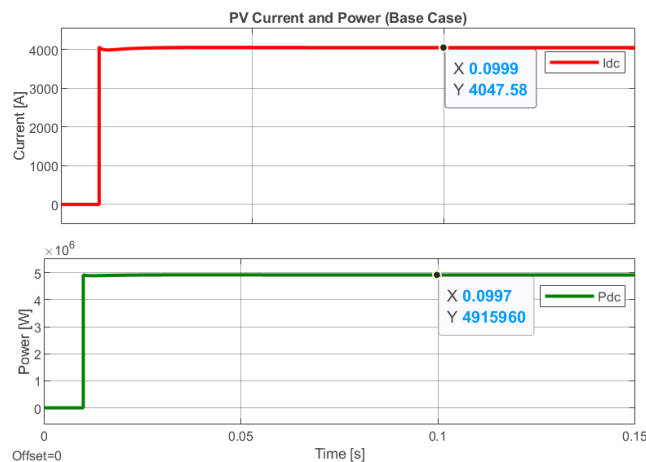


Figure 13: I_{DC} and P_{DC} with Constant G&T

We can see that the steady-state current and power match those calculated for the nominal case once the V_{DC} reference is set properly, validating our approach for the other scenarios.

3.1.2 T Variation

For the next simulation, the temperature of the PV plant drops to 21°C at 0.05 s, after the startup transients have settled. The calculated voltage reference is 1228.4 V, and the other salient features of this disturbance are shown in Fig. 14.

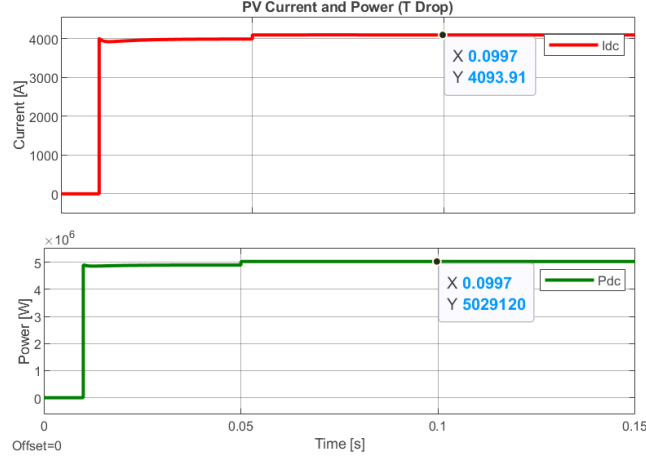


Figure 14: I_{DC} and P_{DC} with Decreased T

The simulation produces the same steady-state current, and therefore power, increase as predicted by the calculation in Fig. 3b. The voltage and current both increase compared to the base case because solar panels are more efficient at lower temperatures.

3.1.3 G Variation

Next, the irradiance is dropped to 600 W/m². The calculated voltage reference for this scenario is 1000.3 V.

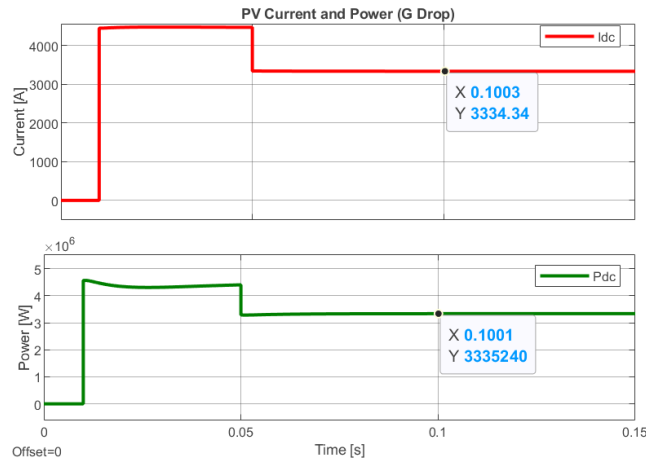


Figure 15: I_{DC} and P_{DC} with Decreased G

Again, the simulation results match those calculated in Fig. 3c. The voltage, current, and power all decrease significantly because G is a quantification of how much energy from the sun is shining on the panels. Therefore, a large drop in production from a PV is expected during a 25% drop in irradiance.

3.1.4 T&G Variation

Lastly, both of the previous variations are combined. The PV voltage is 1001.1 V.

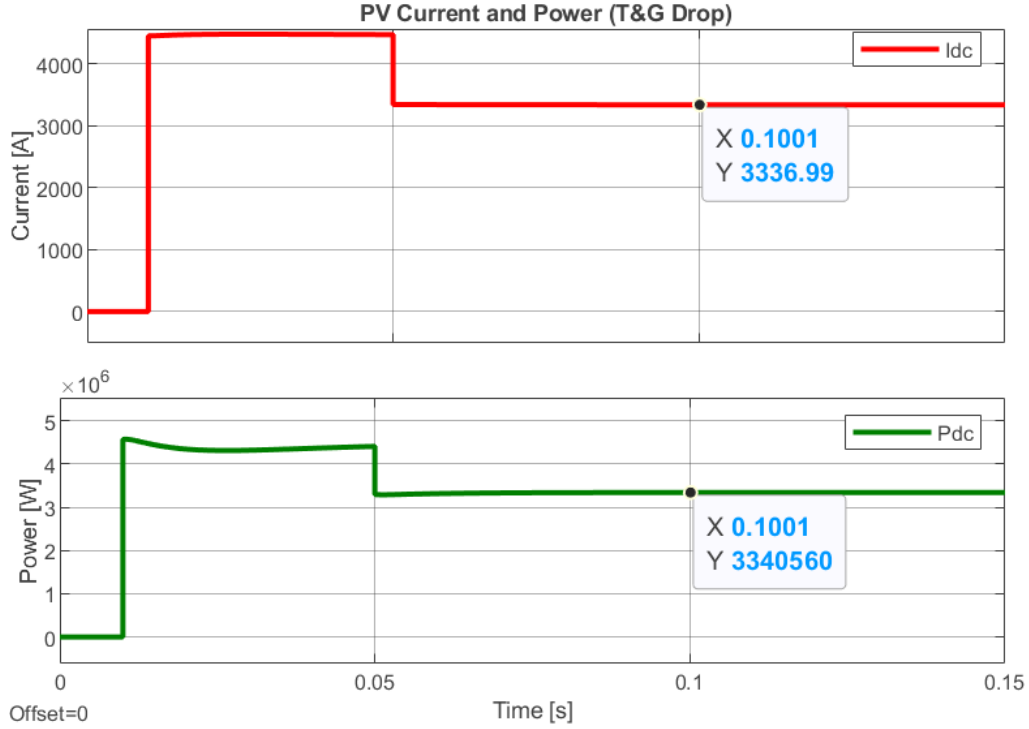


Figure 16: I_{DC} and P_{DC} with Decreased G&T

This simulation largely resembles the previous one as well as matching the expected steady-state of Fig. 3d. This is because change in irradiance affects the PV current more significantly than comparable changes in temperature. So while the voltage does increase slightly compared to the 25°C case, the decreased irradiance causes the overall result to still be a huge reduction in power.

3.2 Reactive Power Reference Variation

The DC voltage controller works by controlling the q-component of the current reference, or the term that controls active power. The reactive power, therefore, is still controlled by the PI implemented in the previous assignment. In the simulation below, the Q reference is changed from 1 to 3 MVAR at 0.1 s to observe the effect on the DC system in the base case.

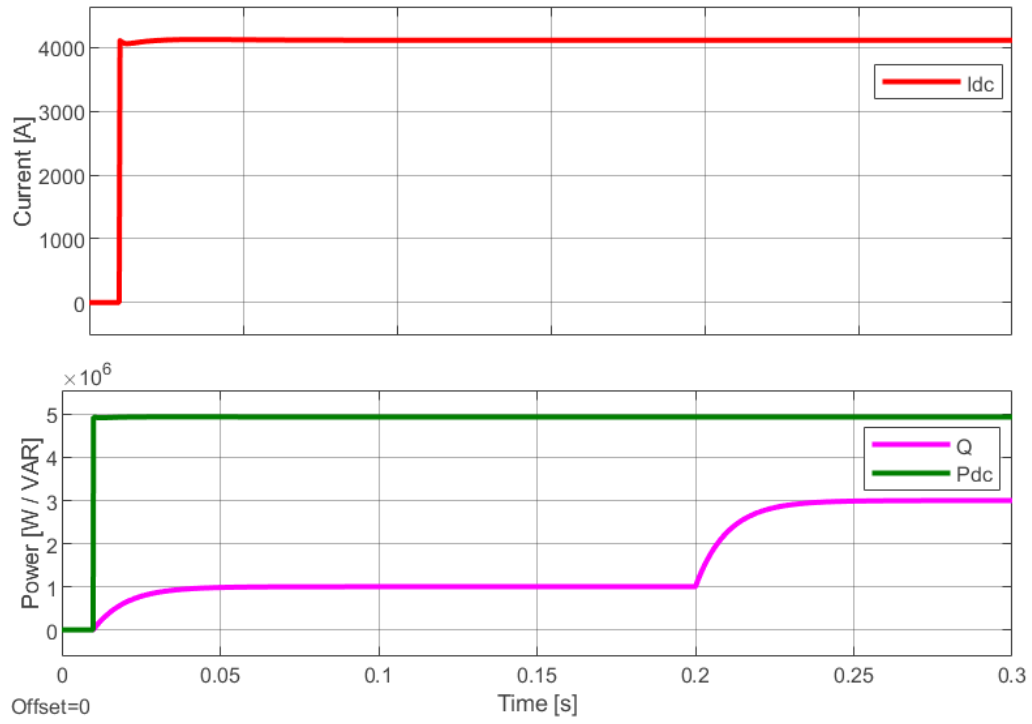


Figure 17: I_{DC} and P_{DC} under Q_{ref} Variation

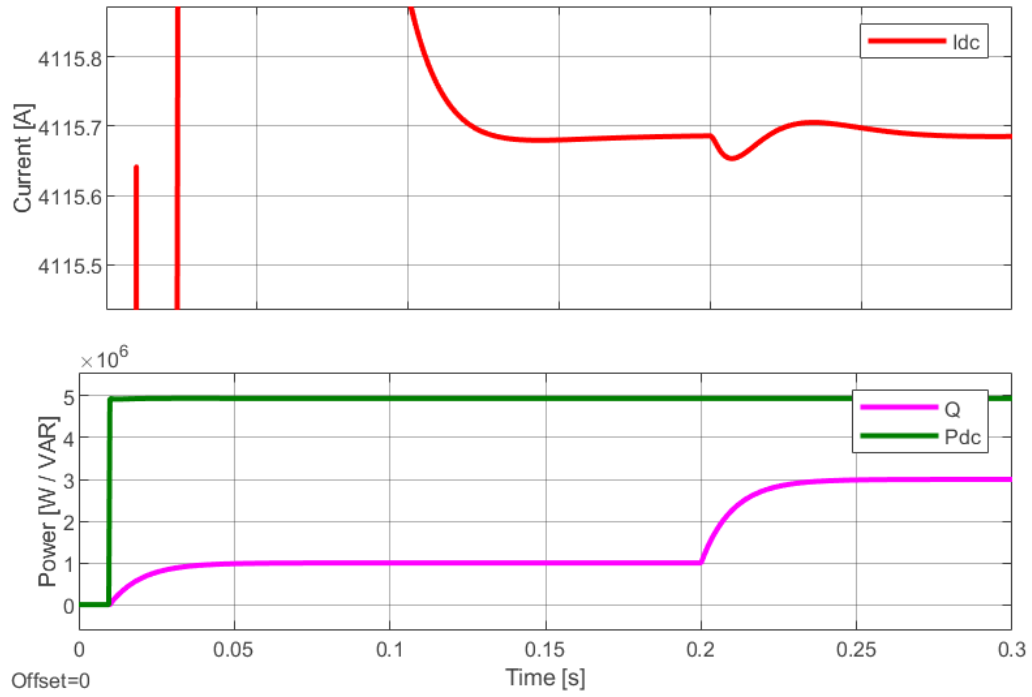


Figure 18: I_{DC} and P_{DC} under Q_{ref} Variation

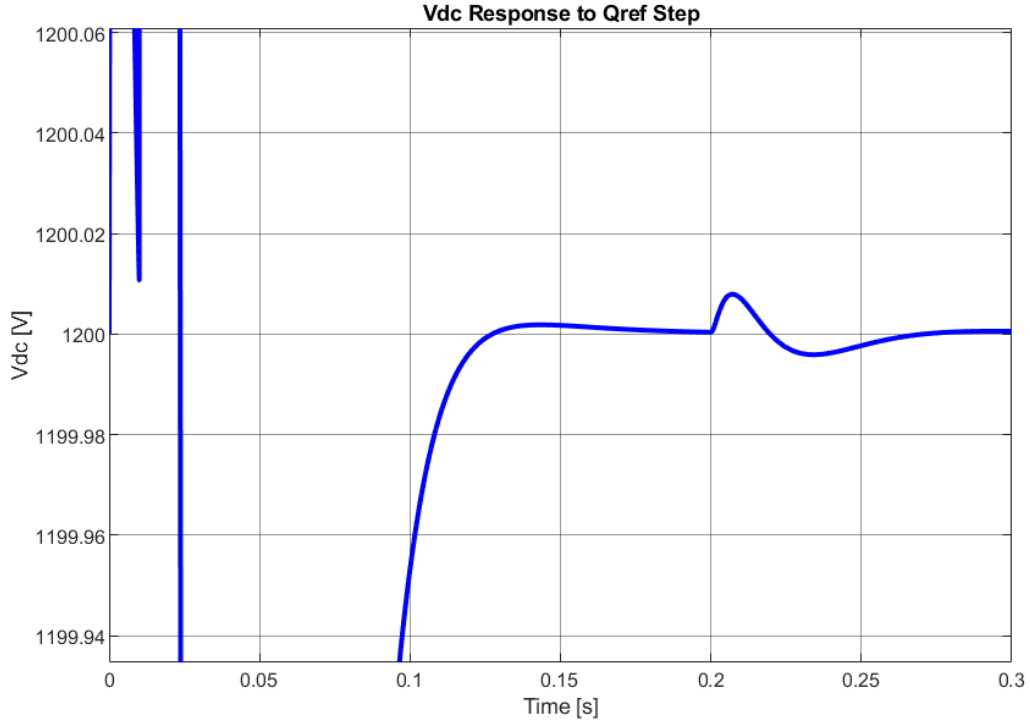


Figure 19: V_{DC} under Q_{ref} Variation

The figures above show the reaction of the system to the reactive power step. While the reactive power output of the inverter increases as expected, the other salient characteristics of the PV array are only impacted slightly. This is because the DC controller and Q controller are responsible for the two different components of the current references in the qd frame corresponding to active and reactive power, respectively. The current reference corresponding to active power does not change significantly with a reactive power step. Figs. 18 and 19 show very small disturbances in reaction to the step in Q_{ref} attributable to error in the decoupling between P and Q. More realistic step changes in reactive power would produce even less noticeable reactions from the PV array.

3.3 Validation of Steady-State Simulation Results

Fig. 20 shows the results in the steady-state calculations performed in MATLAB. The validation covered again three distinct operating scenarios as presented in Section 1.4. The consistency of these results reinforces the model's validity and its applicability for further studies.

	P DC [MW]	Va DC [kV]	Ia DC [kA]
4.9171 MW, -2 MVar, G = 800 W/m ² , T = 25 °C	4.9171	1.2145	4.0486

(a) Irradiance $G = 800 \text{ W/m}^2$ and Temperature $T = 25^\circ\text{C}$

	P DC [MW]	Va DC [kV]	Ia DC [kA]
4.9457 MW, -2 MVar, G = 800 W/m ² , T = 24 °C	4.9457	1.2181	4.0602
4.9741 MW, -1 MVar, G = 800 W/m ² , T = 23 °C	4.9741	1.2216	4.0719
5.0023 MW, 0 MVar, G = 800 W/m ² , T = 22 °C	5.0023	1.225	4.0835
5.0302 MW, 1 MVar, G = 800 W/m ² , T = 21 °C	5.0302	1.2284	4.0949

(b) Irradiance $G = 800 \text{ W/m}^2$ at Different Temperatures T

	P DC [MW]	Va DC [kV]	Ia DC [kA]
4.6312 MW, -2 MVar, G = 750 W/m ² , T = 25 °C	4.6312	1.1786	3.9294
4.2704 MW, -1 MVar, G = 700 W/m ² , T = 25 °C	4.2704	1.1319	3.7728
3.8301 MW, 0 MVar, G = 650 W/m ² , T = 25 °C	3.8301	1.0719	3.5731
3.3356 MW, 1 MVar, G = 600 W/m ² , T = 25 °C	3.3356	1.0003	3.3344

(c) Different Irradiance G and Temperature $T = 25^\circ\text{C}$

	P DC [MW]	Va DC [kV]	Ia DC [kA]
4.6538 MW, -2 MVar, G = 750 W/m ² , T = 24 °C	4.6538	1.1818	3.938
4.2994 MW, -1 MVar, G = 700 W/m ² , T = 23 °C	4.2994	1.1357	3.7857
3.8497 MW, 0 MVar, G = 650 W/m ² , T = 22 °C	3.8497	1.0747	3.5822
3.3408 MW, 1 MVar, G = 600 W/m ² , T = 21 °C	3.3408	1.0011	3.3371

(d) Different Irradiance G and Temperature T

Figure 20: Steady-state PV array results under different irradiance (G) and temperature (T) conditions

4 Task 2.4: Maximum Power Tracking

4.1 Integration and Validation of the MPPT System

To enhance the performance of the PV system under dynamic environmental conditions, a Maximum Power Point Tracking (MPPT) algorithm was integrated into the simulation. The MPPT ensures that the system continuously operates at its optimal power point despite fluctuations in irradiance and temperature. Several methods exist for MPPT, and this implementation leverages the electrical behavior of the PV module, specifically the relationship between voltage (V), current (I), and power (P). The goal is to find the operating point where the power $P = V \times I$ is maximized.

An approximate linear relationship can be used to estimate the maximum power point (MPP) based on either the open-circuit voltage (V_{oc}) or short-circuit current (I_{sc}):

$$V_{mpp} \approx k_v \cdot V_{oc} \quad (18)$$

$$I_{mpp} \approx k_i \cdot I_{sc} \quad (19)$$

Where k_v and k_i are empirical constants that depend on the PV module characteristics. [6] These reference values can be estimated using temperature and irradiance measurements as follows:

$$I_{sc} = I_{sc,n} + K_I(T - T_n) \cdot \frac{G}{G_n} \quad (20)$$

$$V_{oc} = V_{oc,n} + K_V(T - T_n) \quad (21)$$

where:

- $I_{sc,n}$ and $V_{oc,n}$ are the short-circuit current and open-circuit voltage under standard test conditions (STC),
- K_I and K_V are the temperature coefficients of current and voltage,
- T is the current temperature, and T_n is the nominal temperature,
- G is the current irradiance, and G_n is the nominal irradiance.

Fig. 21 shows the complete photovoltaic model with MPPT implementation. The time-domain simulation results for V_{dc} , I_{dc} , and active power are presented in Fig. 22, using the open-circuit voltage method. In this simulation, irradiance and temperature vary over time. As described in the previous section, the irradiance and temperature profiles follow the scenarios outlined below:

- **Constant Irradiance, Decreasing Temperature:**

Fig. 22a and Fig. 22b present the results when irradiance G is held constant at 800 W/m^2 , while temperature decreases progressively by 1°C every 0.03 seconds from 0.06 to 0.18 seconds, ranging from 25°C to 21°C . Under these conditions, voltage V_{dc} , current I_{dc} , and power increase as temperature decreases.

- **Constant Temperature, Decreasing Irradiance:**

Fig. 22c and Fig. 22d show the results when temperature is fixed at 25°C , and irradiance decreases progressively by 50 W/m^2 every 0.03 seconds from 0.03 to 0.18 seconds, ranging from 800 to 600 W/m^2 . In this scenario, current I_{dc} and power P_{dc} decrease, while voltage V_{dc} remains constant, as it depends primarily on temperature, in accordance with (21).

- **Decreasing Both Irradiance and Temperature:**

Fig. 22e and Fig. 22f depict the case where both irradiance and temperature decrease simultaneously, using the same values as in the previous scenarios. In this case, voltage V_{dc} increases, while current and power decrease.

Compared to the fixed-voltage system without MPPT in Section 3, the MPPT enabled system achieves better power extraction and more stable voltage and current profiles. Without MPPT, the system maintains a constant voltage that fails to track the shifting maximum power point caused by changes in temperature and irradiance, leading to suboptimal performance. In contrast, MPPT controller dynamically adjusts the operating point in real time, ensuring maximum energy conversion efficiency under varying environmental conditions.

Note: The initial peak transient observed in V_{dc} is due to the initial charge of the capacitor. This does not significantly affect the overall results of the simulation.

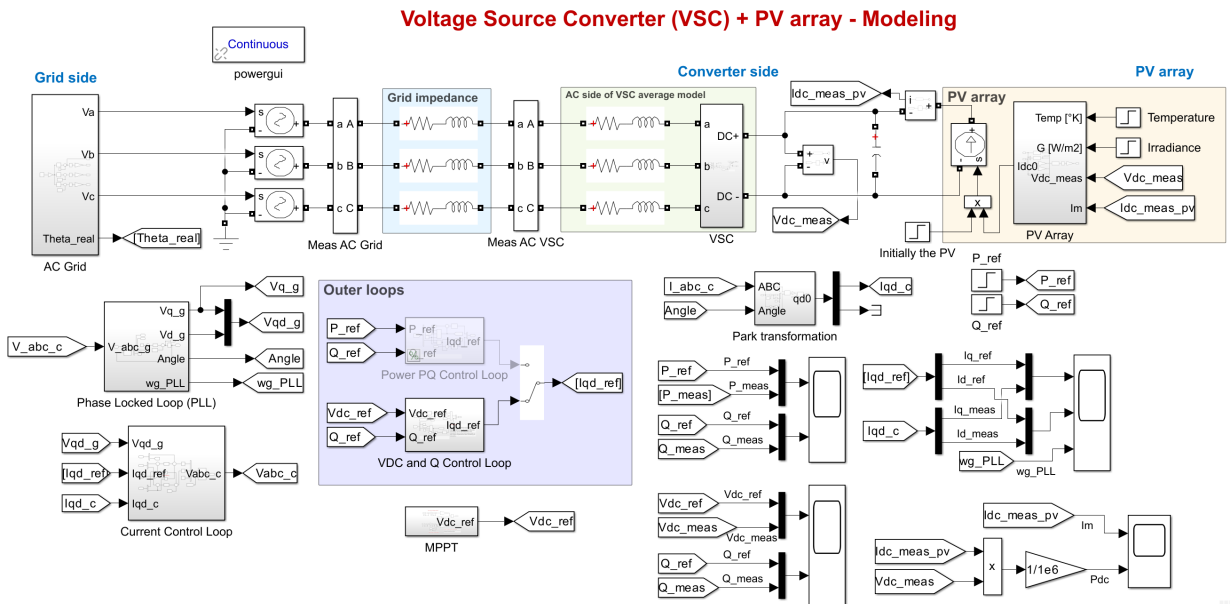


Figure 21: Complete photovoltaic system model with MPPT implementation

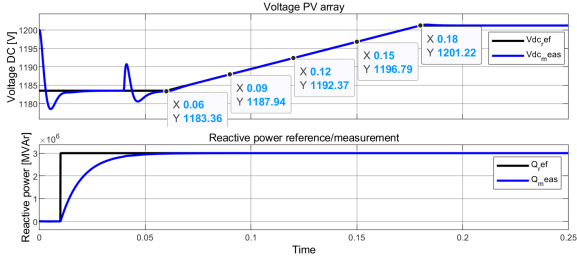
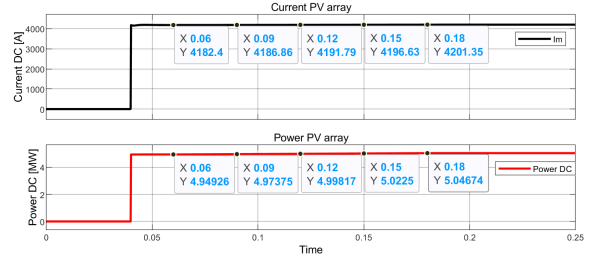
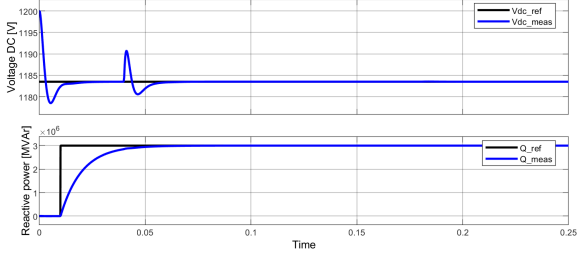
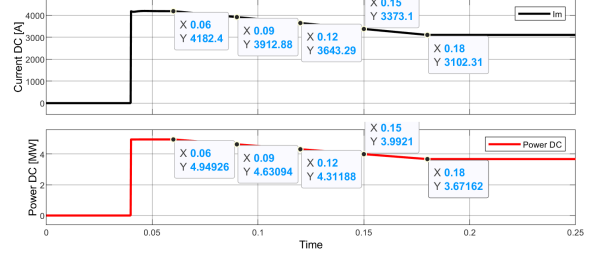
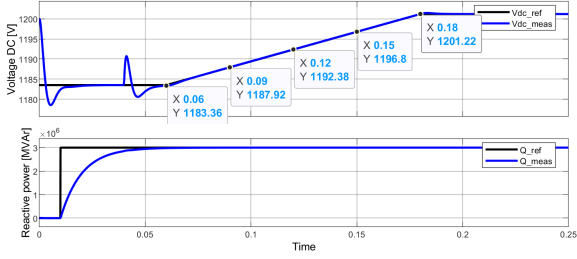
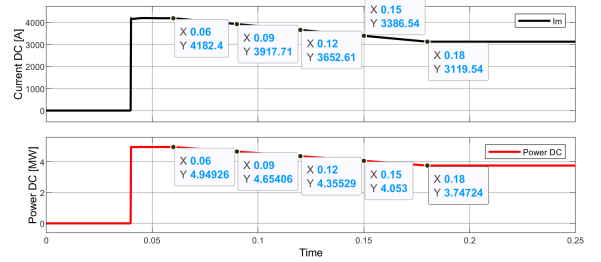

 (a) Vdc and reactive power results at $G = 800 \text{ W/m}^2$ and different T

 (b) Ia and Power at $G = 800 \text{ W/m}^2$ and different T

 (c) Vdc and reactive power results at different G and $T = 25^\circ\text{C}$

 (d) Ia and Power at different G and $T = 25^\circ\text{C}$

 (e) Vdc and reactive power results at different G and T

 (f) Ia and Power at different G and T

 Figure 22: Dynamic behavior of the PV array under different irradiance (G) and temperature (T) conditions using the MPPT method

4.2 Validation of Steady-State Simulation Results

The steady-state results from the MPPT-enhanced Simulink model were validated against the theoretical values calculated in MATLAB. Three operating scenarios were analyzed: (1) constant irradiance with decreasing temperature, (2) constant temperature with decreasing irradiance, and (3) simultaneous variation of both. In all cases, key electrical parameters V_{dc} , I_{dc} , and output power showed strong agreement between dynamic simulations (Fig. 22) and the MATLAB-based steady-state calculations (Fig. 23), validating the accuracy of the model. The AC active power is assumed equal to the DC active power.

Additionally, the results incorporate both direct methods discussed earlier for determining the maximum power point. The output power for both approaches is identical, confirming the correct implementation of the MPPT algorithm. Furthermore, it is important to highlight that in the open-circuit voltage method, the operating voltage $V_{av} = V_{dc}$ remains constant when

the temperature T is fixed, as illustrated in Fig. 23c and explained in (21).

	Open-Circuit Voltage Method			Open-Circuit Current Method		
	P_v DC [MW]	Va_v DC [kV]	Ia_v DC [kA]	P_i DC [MW]	Va_i DC [kV]	Ia_i DC [kA]
4.9493 MW, -2 MVar, $G = 800 \text{ W/m}^2$, $T = 25 \text{ }^\circ\text{C}$	4.9493	1183.5	4181.9	4.949	1178.1	4200.7

(a) Irradiance $G = 800 \text{ W/m}^2$ and Temperature $T = 25^\circ\text{C}$

	Open-Circuit Voltage Method			Open-Circuit Current Method		
	P_v DC [MW]	Va_v DC [kV]	Ia_v DC [kA]	P_i DC [MW]	Va_i DC [kV]	Ia_i DC [kA]
4.9738 MW, -2 MVar, $G = 800 \text{ W/m}^2$, $T = 24 \text{ }^\circ\text{C}$	4.9738	1187.9	4186.9	4.9735	1184.4	4199.1
4.9982 MW, -1 MVar, $G = 800 \text{ W/m}^2$, $T = 23 \text{ }^\circ\text{C}$	4.9982	1192.3	4191.9	4.998	1190.7	4197.4
5.0225 MW, 0 MVar, $G = 800 \text{ W/m}^2$, $T = 22 \text{ }^\circ\text{C}$	5.0225	1196.8	4196.7	5.0225	1197	4195.8
5.0467 MW, 1 MVar, $G = 800 \text{ W/m}^2$, $T = 21 \text{ }^\circ\text{C}$	5.0467	1201.2	4201.4	5.047	1203.3	4194.2

(b) Irradiance $G = 800 \text{ W/m}^2$ at Different Temperatures T

	Open-Circuit Voltage Method			Open-Circuit Current Method		
	P_v DC [MW]	Va_v DC [kV]	Ia_v DC [kA]	P_i DC [MW]	Va_i DC [kV]	Ia_i DC [kA]
4.6309 MW, -2 MVar, $G = 750 \text{ W/m}^2$, $T = 25 \text{ }^\circ\text{C}$	4.6309	1183.5	3912.9	4.6308	1175.9	3938.2
4.3119 MW, -1 MVar, $G = 700 \text{ W/m}^2$, $T = 25 \text{ }^\circ\text{C}$	4.3119	1183.5	3643.3	4.3121	1173.2	3675.6
3.9921 MW, 0 MVar, $G = 650 \text{ W/m}^2$, $T = 25 \text{ }^\circ\text{C}$	3.9921	1183.5	3373.1	3.9928	1169.9	3413.1
3.6716 MW, 1 MVar, $G = 600 \text{ W/m}^2$, $T = 25 \text{ }^\circ\text{C}$	3.6716	1183.5	3102.3	3.6732	1165.9	3150.5

(c) Different Irradiance G and Temperature $T = 25^\circ\text{C}$

	Open-Circuit Voltage Method			Open-Circuit Current Method		
	P_v DC [MW]	Va_v DC [kV]	Ia_v DC [kA]	P_i DC [MW]	Va_i DC [kV]	Ia_i DC [kA]
4.6541 MW, -2 MVar, $G = 750 \text{ W/m}^2$, $T = 24 \text{ }^\circ\text{C}$	4.6541	1187.9	3917.8	4.6539	1182.2	3936.6
4.3553 MW, -1 MVar, $G = 700 \text{ W/m}^2$, $T = 23 \text{ }^\circ\text{C}$	4.3553	1192.3	3652.7	4.3552	1185.8	3672.8
4.0530 MW, 0 MVar, $G = 650 \text{ W/m}^2$, $T = 22 \text{ }^\circ\text{C}$	4.053	1196.8	3386.6	4.053	1188.9	3409.1
3.7472 MW, 1 MVar, $G = 600 \text{ W/m}^2$, $T = 21 \text{ }^\circ\text{C}$	3.7472	1201.2	3119.6	3.7475	1191.3	3145.6

(d) Different Irradiance G and Temperature T

Figure 23: Steady-state PV array results under different irradiance (G) and temperature (T) conditions using the MPPT method

5 Conclusions

- Temperature and irradiance are critical factors influencing the electrical output of a photovoltaic (PV) panel. An increase in irradiance results in a significant rise in current generation, as demonstrated in Fig. 2c and Fig. 3c. Conversely, lower temperatures enhance the voltage output but lead to a slight decrease in current.
- The cascaded PI-based DC-voltage controller, with gains set by

$$k_{p_v} = 2\xi\omega_n C_{dc}, \quad k_{i_v} = \omega_n^2 C_{dc},$$

ensures the DC bus voltage tracks its reference without steady-state error while remaining slower than the inner current loop to preserve stability. Adding a feedforward term as seen in leverages the measured disturbance current to sharply reduce voltage spikes under load steps as in seen in Fig. 6 through to Fig. 10. However, without it, faster PI tuning yields quicker settling but larger peaks. The best compromise is critical damping ($\xi \approx 1$), moderate ω_n , and feedforward for fast, low-overshoot, disturbance-resilient control this were demonstrated in the Figs. 11 and 12 respectively.

- Changing the reactive power reference causes the i_d^* to adjust accordingly, while the DC voltage (V_{dc}), DC current (I_{dc}), and active power from the PV array remain largely unaffected. This is because active and reactive power are controlled independently through the q - and d -components of the current reference. Minor disturbances observed in i_q when i_d is changed are due to imperfect decoupling and delays in the current control loop. This delay is intentionally introduced to avoid algebraic errors that can occur when the system directly uses output values as reference. Overall, the active power path remains stable despite reactive power variations.
- Variations in voltage and current due to environmental conditions directly affect the Maximum Power Point (MPP) of the PV system. Therefore, implementing an efficient Maximum Power Point Tracking (MPPT) algorithm becomes essential to dynamically adjust the operating point and maximize power extraction under changing irradiance and temperature conditions. The results shown in Fig. 22 and Fig. 23 reinforce the importance of adaptive MPPT strategies to maintain optimal energy yield and system efficiency in real-world applications.

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